



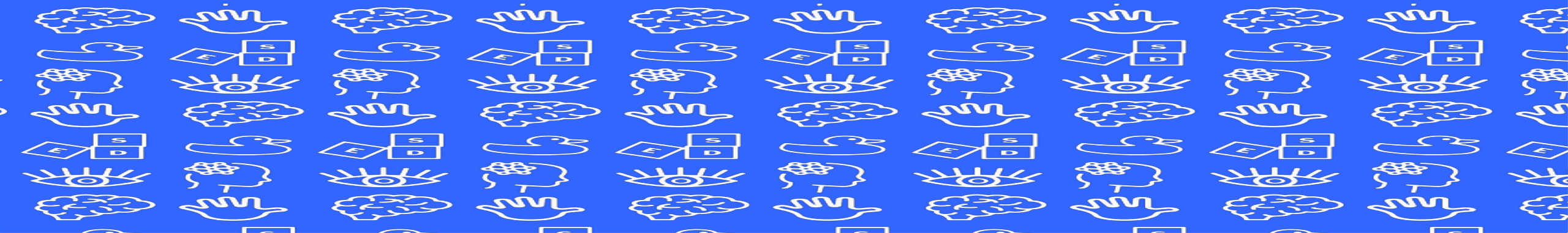
South Carolina

Caregiver Acceptability Clusters and Data-Trust Screening

Infant Autism Screening survey · REDCap projects 4797 and 4581

Early Social Development Lab · University of South Carolina

Verified cache snapshot 2026-08-14 · 1,956 cached records · Prepared for PI, clinical and technical review



One survey, two questions, one held-out design

Profiles are built only from Thoughts, Feelings and Attitudes items, and every outcome we care about is held out of that construction.

1,956

cached records across both projects

131

caregivers assigned to a profile

127

with a valid held-out screening outcome

10

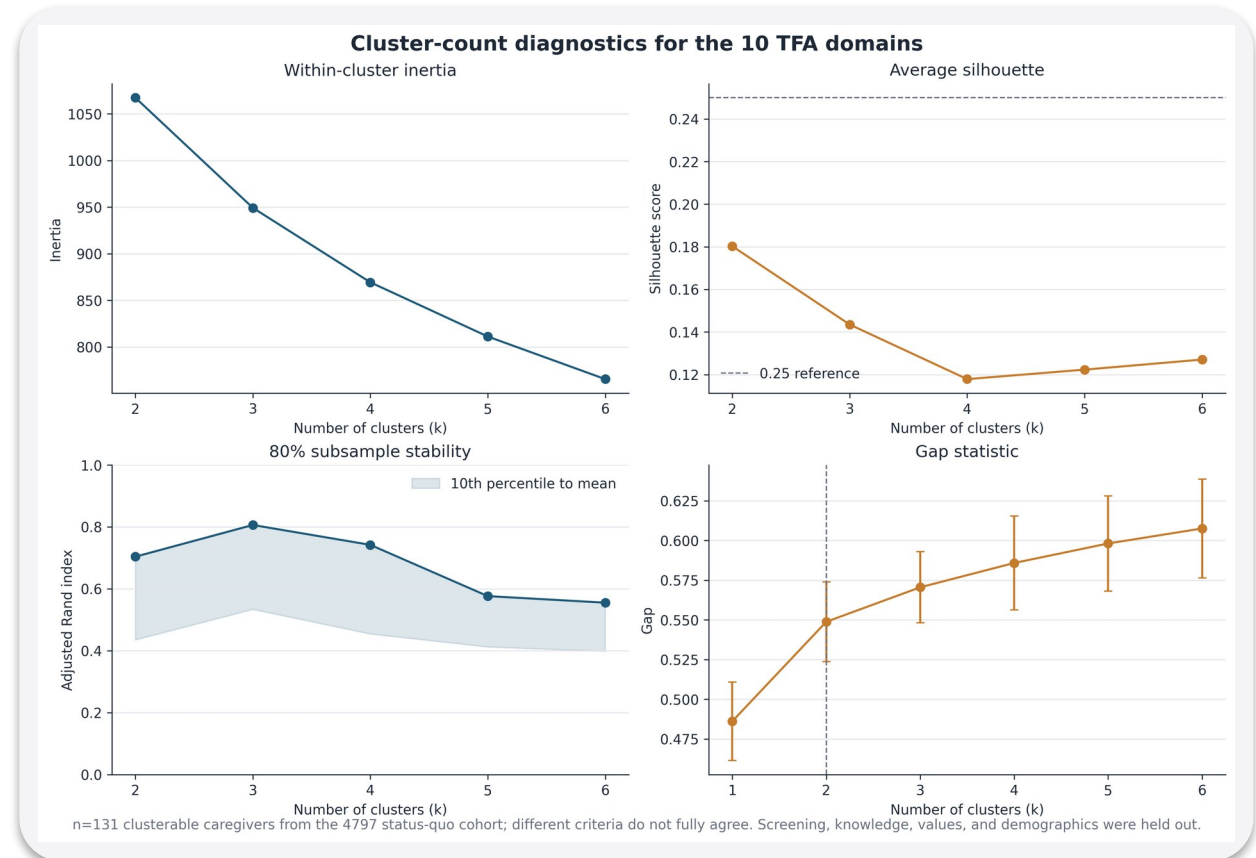
deterministic trust rules, R1 to R10

- The 1,956 records combine clean_4797 (n=177) and dirty_4581 (n=1,779); UIDs are namespaced so that 174 overlapping numeric record IDs cannot corrupt joins
- Clustering uses the 10 TFA domains only. Screening intent, autism knowledge, values and demographics are held out and examined only after profile assignment
- Cohort flow: 135 selected analytic records, 131 cluster-assigned at a threshold of at least 8 of 10 input domains, 127 of those with a valid screening outcome
- The trust screen is deterministic and rule-based. It runs on all 1,956 records, and no outcome variable enters any rule or any tier

Two profiles is the defensible choice, and the separation is modest

Selection criteria disagree, so $k=2$ was chosen for stability and interpretability, not because a clean natural break exists.

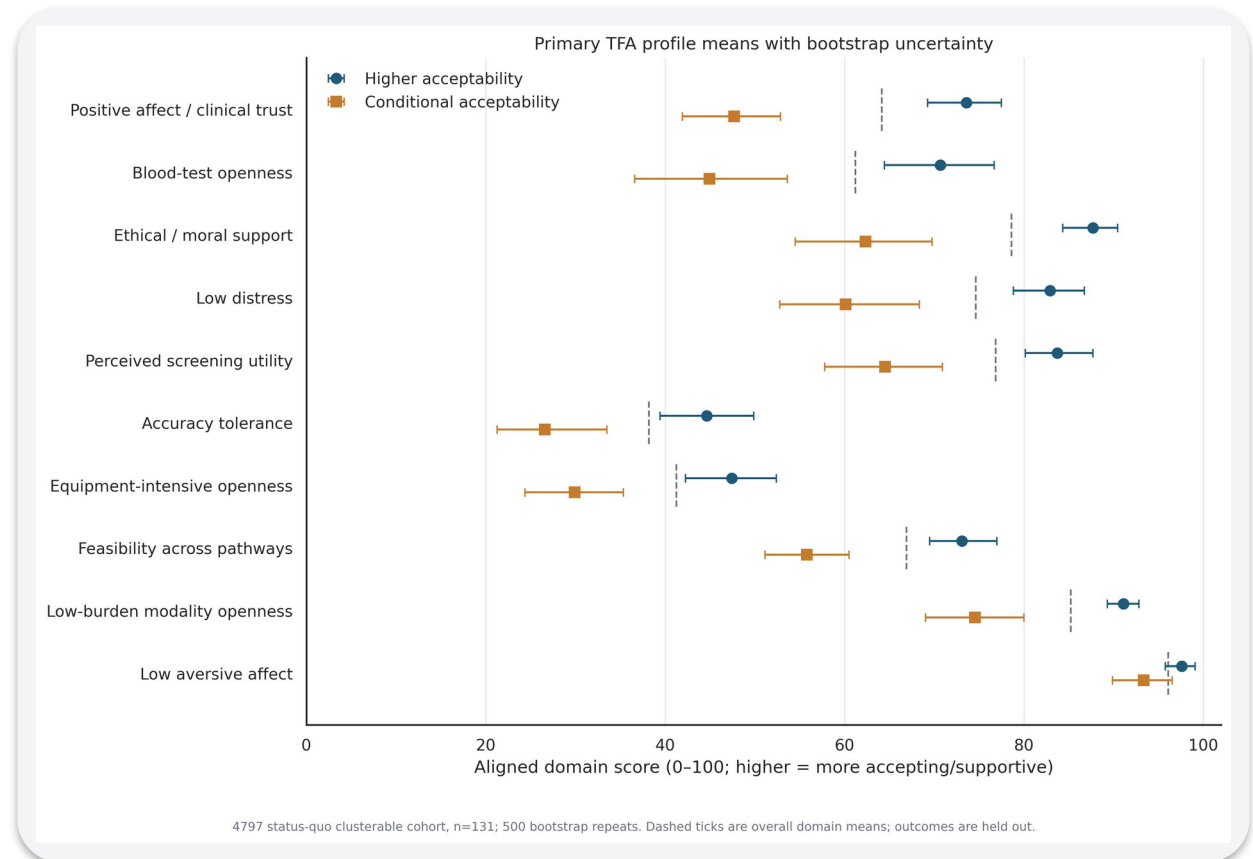
- Average silhouette peaks at $k=2$ (0.180) but stays below the 0.25 reference at every k tested
- 80% subsample stability is 0.704 ARI at $k=2$ and higher at $k=3$ (0.806), but $k=3$ places only 9 caregivers, 6.9% of the cohort, in its smallest cluster
- The gap statistic rises monotonically through $k=6$, so it never nominates an interior optimum
- Treat the result as a partition of a gradient rather than the discovery of two discrete caregiver types



Trust, blood-test openness and ethical support separate the profiles

The two profiles differ most on relational and ethical dimensions, and barely at all on how aversive caregivers find the screen.

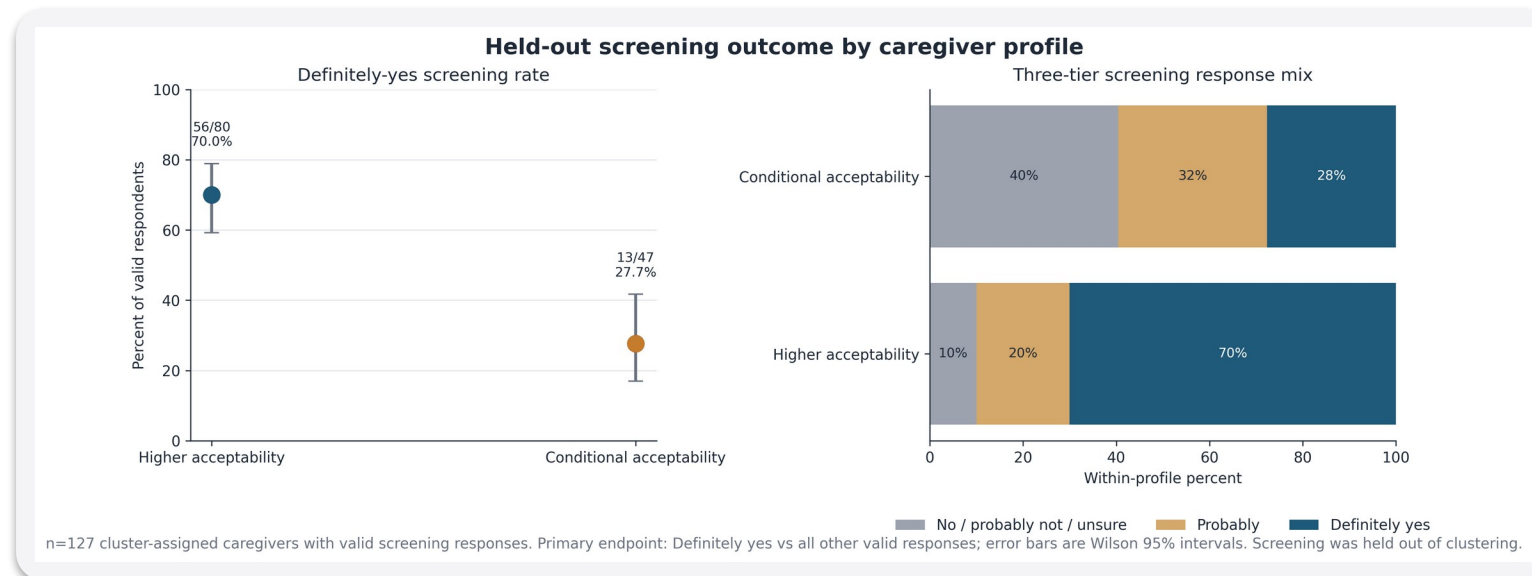
- Higher acceptability n=84 (64.1%), Conditional acceptability n=47 (35.9%); mean TFA acceptability 78.7 versus 59.2 on a 0 to 100 scale
- Largest separations: positive affect and clinical trust 25.9 points, blood-test openness 25.7 points, ethical and moral support 25.3 points
- Smallest separation: low aversive affect, 97.6 versus 93.4, a 4.2-point difference — neither profile finds the screen distressing in itself
- Both profiles sit low on accuracy tolerance (44.6 versus 26.6) and equipment-intensive openness (47.4 versus 29.9)



Screening intent tracks profile membership

Definitely-yes screening intent is 70.0% in the higher-acceptability profile and 27.7% in the conditional profile.

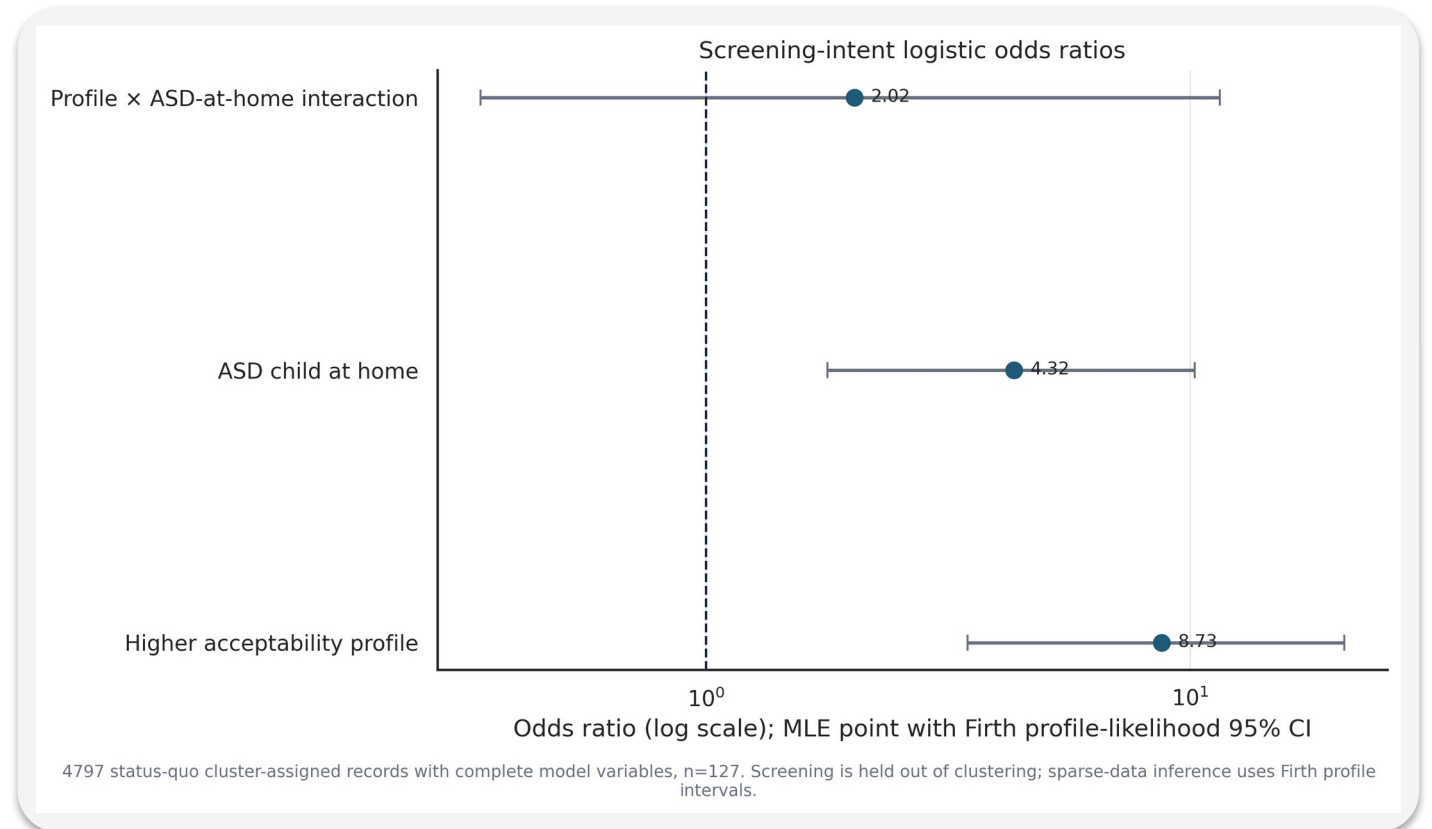
- 56 of 80 versus 13 of 47 answered Definitely yes, a gap of 42.3 percentage points
- Wilson 95% intervals are 59.2 to 78.9% and 16.9 to 41.8%, and they do not overlap
- Widening the endpoint to probably or definitely yes gives 90.0% versus 59.6%, so the ordering is not an artefact of where the cut is placed
- Screening was held out of clustering, so this is an association between an unsupervised partition and an outcome, not a causal effect of profile membership



The profile signal survives adjustment for autism in the household

Profile membership and lived autism experience act additively, and there is no evidence that they interact.

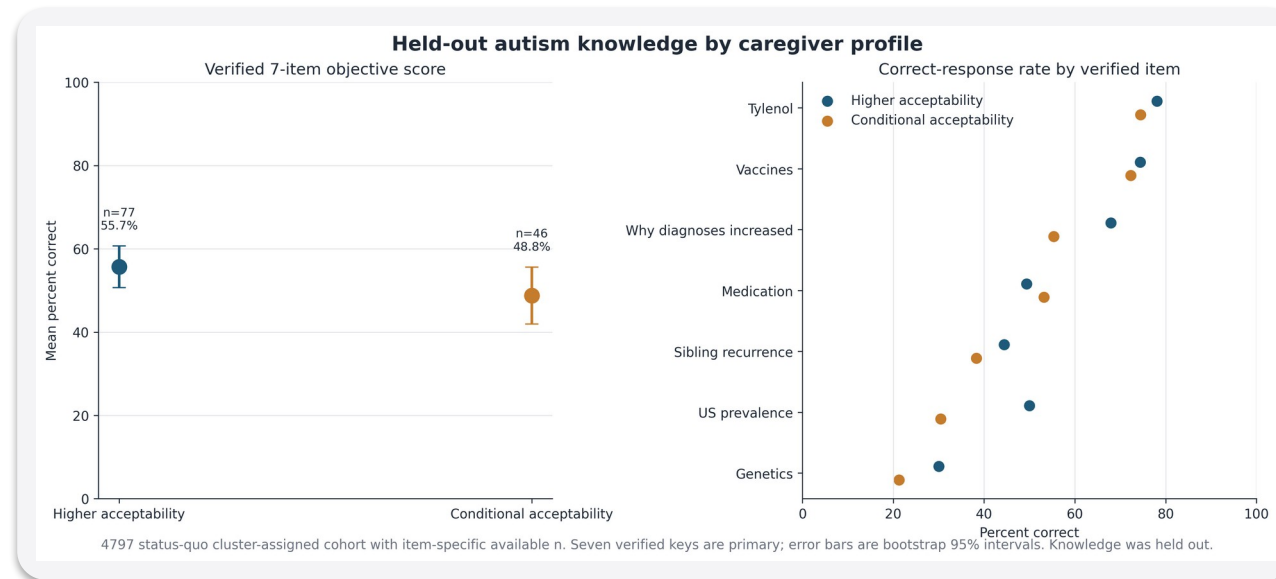
- Higher-acceptability profile adjusted odds ratio 8.73; autistic child at home adjusted odds ratio 4.32 ($p=0.0012$), $n=127$
- The profile-by-ASD interaction term is 2.02 with $p=0.445$, which is not distinguishable from no interaction at this sample size
- The profile gap holds inside both strata: 88.2% versus 37.5% where there is an autistic child at home, and 56.5% versus 17.4% where there is not
- Firth profile-likelihood intervals are used throughout because several cells are sparse and standard maximum likelihood is unstable there



Knowledge, values and demographics do not reliably separate the profiles

After multiplicity control none of the three held-out characterization families distinguishes the two profiles.

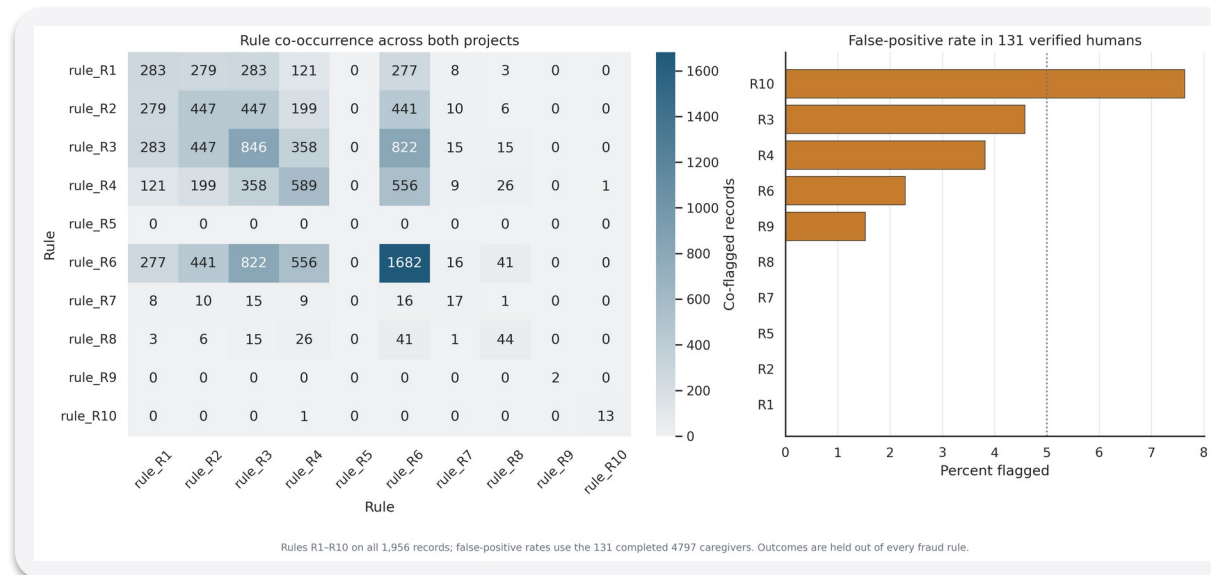
- Verified 7-item autism knowledge is 3.90 versus 3.41 correct (55.7% versus 48.8%), Mann-Whitney $p=0.099$
- Self-direction shows the largest raw values signal (Cohen's d 0.48, $p=0.013$) but its FDR-adjusted p is 0.129
- Every demographic contrast has FDR p at or above 0.56; the closest is any premature birth at raw $p=0.051$, FDR $p=0.56$
- At 47 and 80 respondents the study can only detect roughly a 25.6 point difference at 80% power, so this is absence of evidence, not evidence of equivalence



A graded, deterministic screen — not a bot label

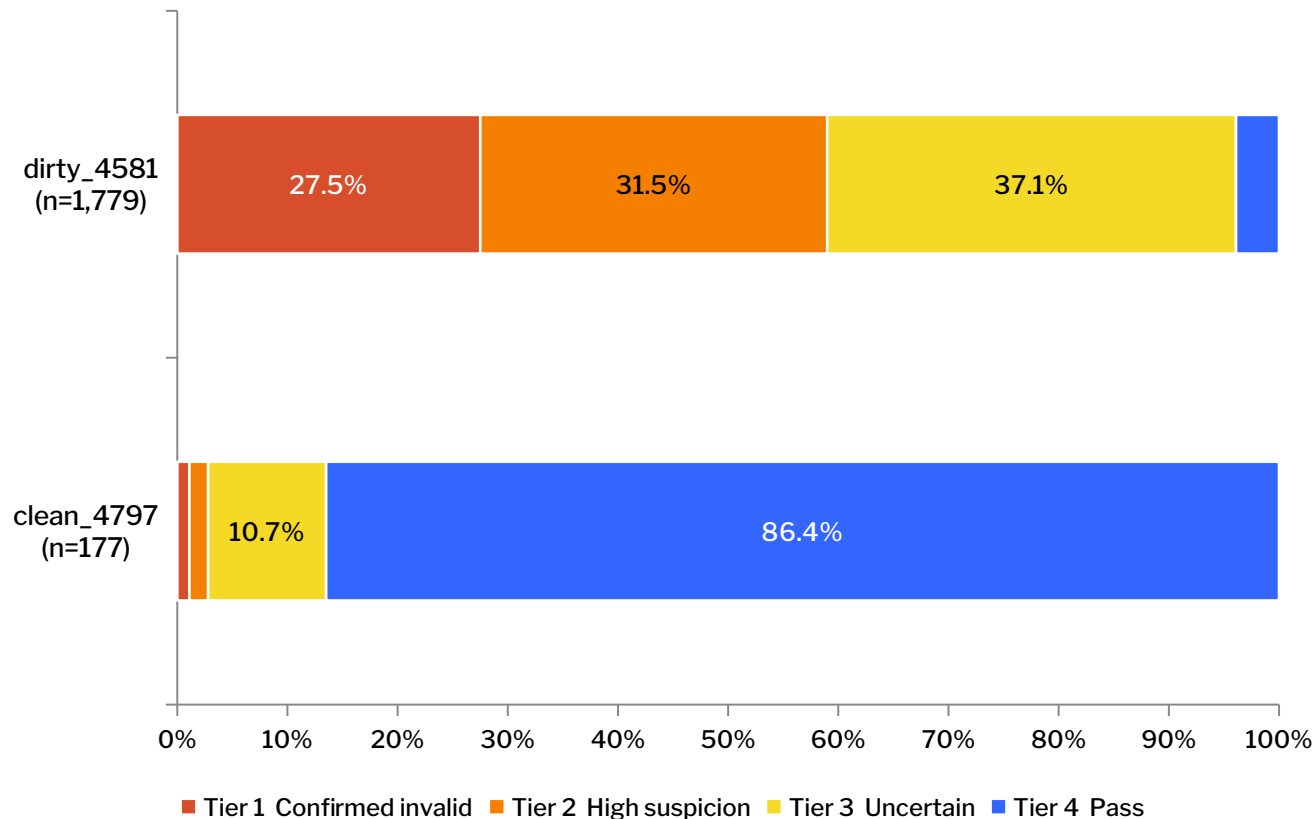
Ten rules anchored to 131 verified-human completions assign every record a tier, and no record is ever labelled a bot.

- Rules R1 to R10 run on all 1,956 records; thresholds are derived from the completed 4797 caregivers and never from any outcome
- False-positive rates measured in verified humans: R10 7.6%, R3 4.6%, R4 3.8%, R6 2.3%, R9 1.5%. R1, R2, R5, R7 and R8 flag nobody
- R6, rapid submission clustering, dominates co-occurrence at 1,682 records, so it is the workhorse and also the least specific rule
- R5, the exact ordered Likert fingerprint duplicate, never fires anywhere in the 1,956 records
- R10 is validation-only and does not define a tier; its email anti-fraud component is unavailable under current API permissions



Pass rates differ sharply between the two projects

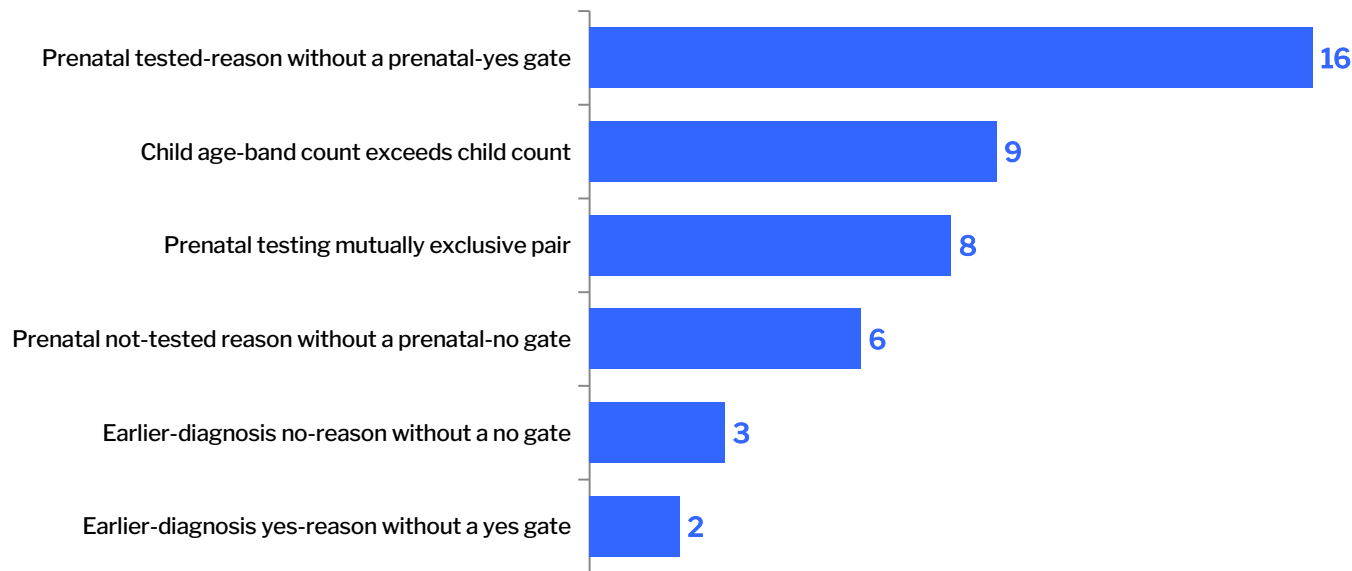
86.4% of clean-project records reach Tier 4, versus 3.9% in the legacy project.



- clean_4797: 153 pass, 19 uncertain, 3 high suspicion, 2 confirmed invalid
- dirty_4581: 70 pass, 660 uncertain, 560 high suspicion, 489 confirmed invalid
- Tiers 1 to 3 account for 96.1% of the legacy project and 13.6% of the clean project
- This describes how much of each project our rules will trust. It is not an estimate of how many respondents were automated

R8 generalized: 44 branching and logic impossibilities

The branching audit turns REDCap skip-logic contradictions into auditable, record-level evidence with zero false positives among verified humans.

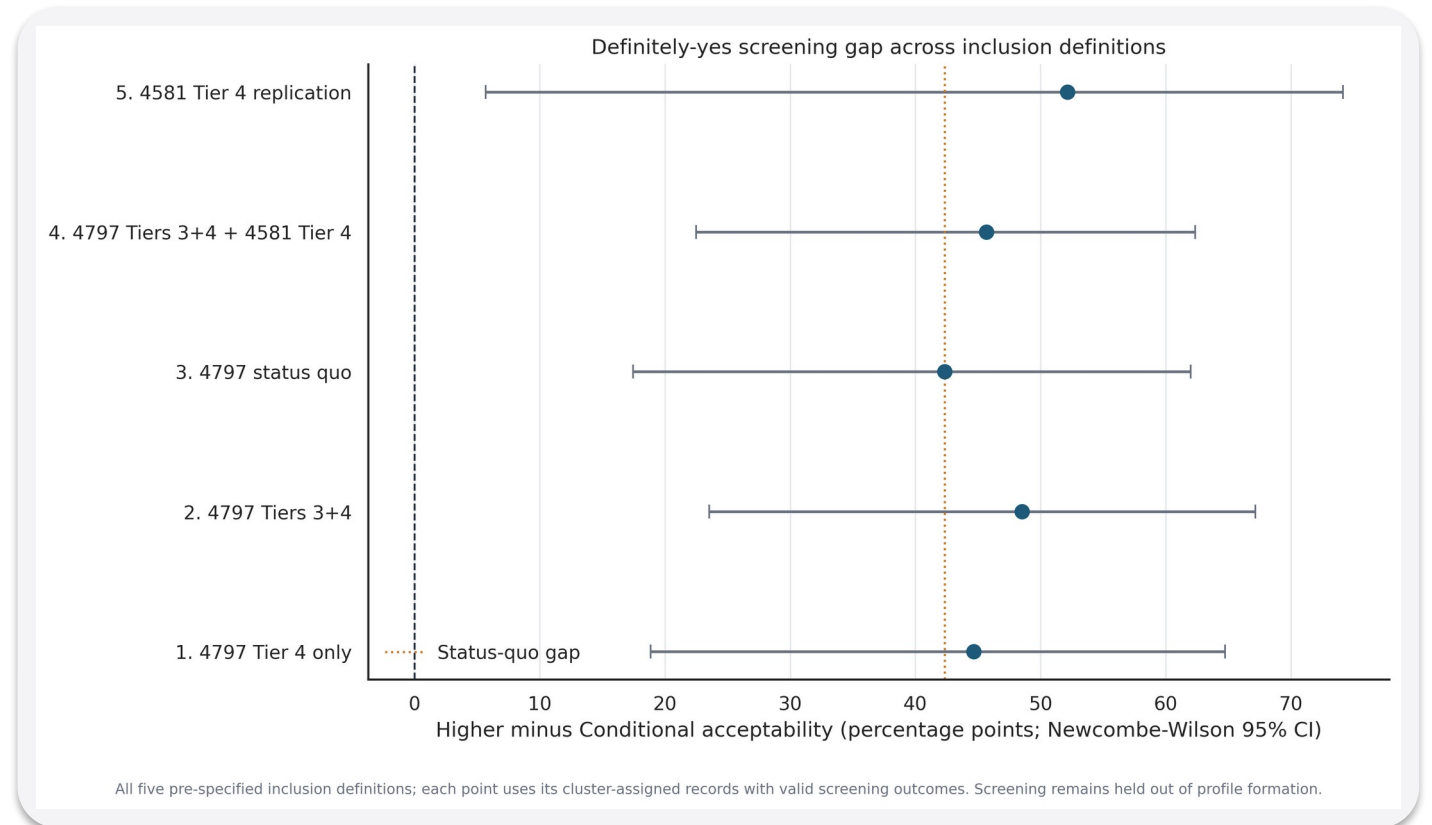


- R8 now covers three families: autistic-child follow-ups that persist when the reported count is zero, prenatal and earlier-diagnosis branching contradictions, and an impossible child age-band count
- 44 flagged records, all 44 in dirty_4581 and 0 in clean_4797, with a false-positive rate of 0.0% in the 131-record verified-human reference
- These are hard logical impossibilities rather than soft suspicion signals, so they stay in Tier 1 and are reviewable field by field against the instrument's own skip logic

Trust filtering does not overturn the profile finding

Across all five pre-specified inclusion definitions the screening gap stays between 42.3 and 52.2 percentage points.

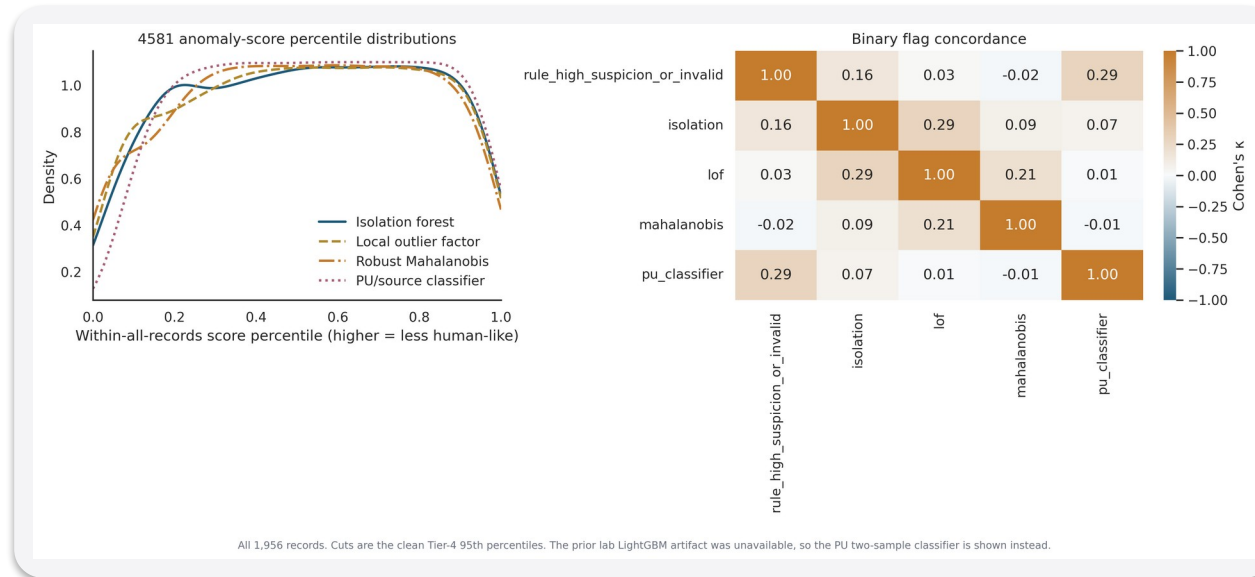
- Tier-4 only 44.7 pp, Tiers 3+4 48.5 pp, status quo 42.3 pp, pooled with 4581 Tier 4 45.7 pp, and 4581 Tier-4 replication 52.2 pp
- Every interval excludes zero, but the 4581 replication is imprecise at 5.7 to 74.1 pp on only 47 respondents
- Firth odds ratios across the five definitions range from 7.39 to 11.09
- Within the primary cohort only 14 of 131 clusterable caregivers sit below Tier 4, so trust filtering has limited room to move this result
- The pooled definition agrees only moderately with the status quo partition (ARI 0.52), so its point estimate is a different analysis, not a confirmation



Independent detectors agree weakly, so no single score is ground truth

Rule flags and unsupervised detectors overlap far less than a reliable detector would, which is the main reason we report tiers rather than labels.

- Best agreement is Cohen's kappa 0.29, between rule flags and the PU/source classifier; isolation forest and local outlier factor also reach only 0.29
- Robust Mahalanobis is essentially unrelated to the rules (kappa -0.02) and to the PU classifier (-0.01)
- The 4797-versus-4581 source classifier reaches AUC 0.980, but the negative control on random clean halves sits at 0.525, so the separation is real yet not attributable to fraud alone
- The strongest discriminating feature is the count of filled open-text fields (permutation AUC drop 0.204), a plausible instrument-design difference rather than a bot signature
- The prior lab LightGBM artifact was not present in the scoped project files, so the PU classifier stands in for it



What we conclude, and what we will not claim

Keep 4797 as the primary cohort, report graded tiers rather than labels, and do not publish a bot prevalence.

	DECISION	RECOMMENDATION	STATUS
D1	Primary cohort	Keep 4797 primary; 4581 is a caveated replication	Implemented
D2	Trust target	Four graded inclusion tiers, not a binary bot label	Implemented
D3	Contamination fraction	Do not present an identifiable bot prevalence	Open, not identifiable
D4	Knowledge scoring	Binary verified correct-count is primary; graded is sensitivity only	Pre-specified in config
D5	Four excluded records	Report the full tipping-point range, 39.0 to 44.5 pp	Implemented

The PU model-implied nonhuman share in 4581 is 94.4% (93.9 to 95.0), but recruitment-period and instrument drift violate the identifying assumption, so it is a separation fraction and not a bot prevalence. It should not be quoted as one.

PART B

Technical appendix

Methods, assumptions, sensitivity checks, detector caveats, branching-logic audit detail and limitations. Eight slides for technical reviewers.



South Carolina

Provenance, caching and the gates that halt the pipeline

Every input is hashed, dated and count-checked, and the pipeline halts rather than silently analysing a drifted cohort.

Hashed, dated caches

4797 record 177×385, 4581 record 1779×369, plus metadata (324 and 315 fields) and instruments. Each Parquet cache carries a SHA-256 sidecar and is dated 2026-08-14.

Hard halt conditions

The run aborts if the 177 / 1,779 record counts change, or if the verified-human timing reference moves off 131 completed 4797 records.

Field alignment

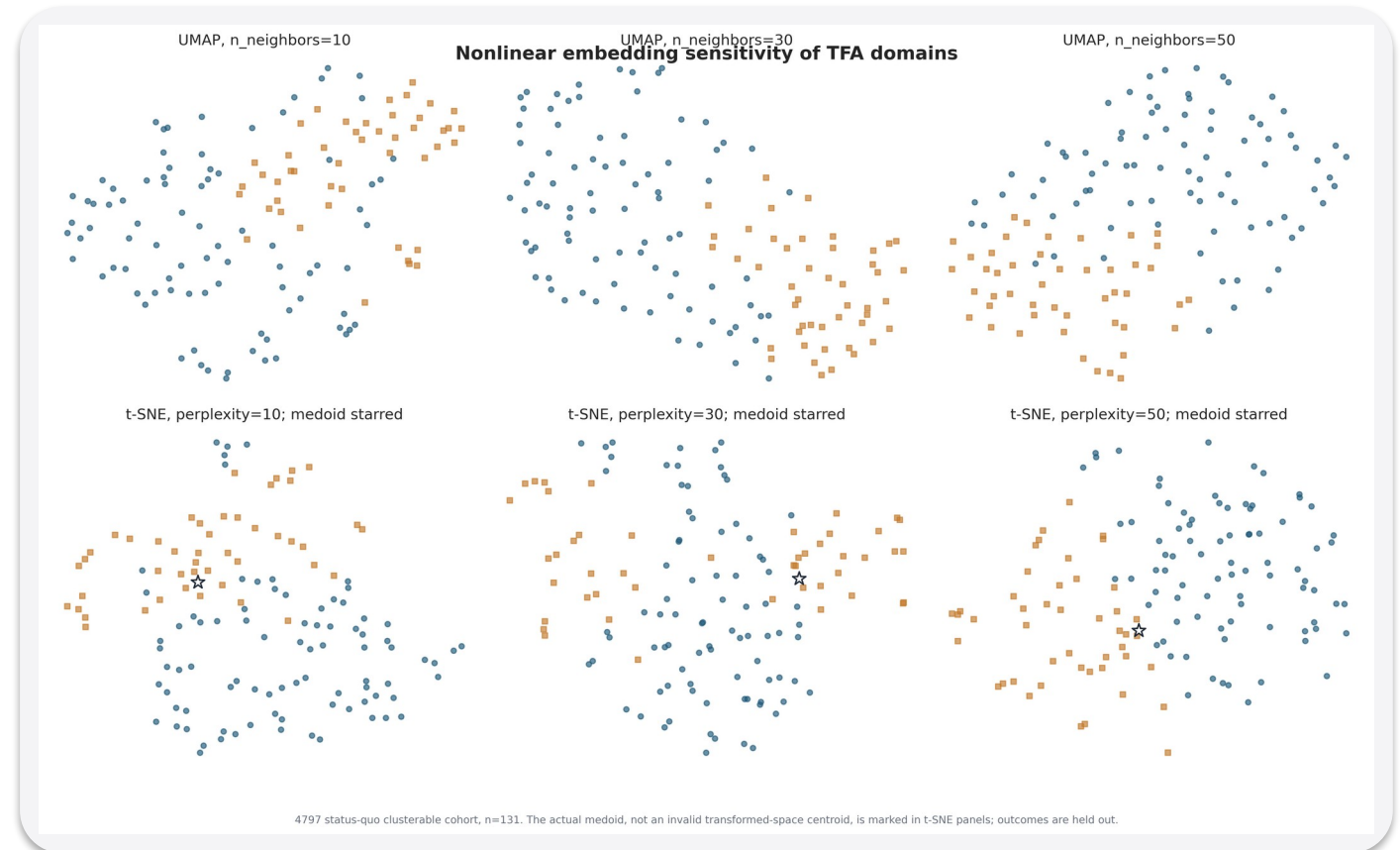
311 fields exist in both projects, 13 only in 4797 and 4 only in 4581. Pooled models use shared behavioural and TFA fields only.

- Namespaced UIDs make all 1,956 records unique despite 174 colliding numeric record IDs across the two projects
- The 135-ID analytic cohort comes from cleaned_autism_study_data.csv; table_1 records that its selection rule is undocumented in the scoped files
- Aggregate outputs carry no email, ZIP, date of birth, occupation text or API token; record_flags.parquet and data_cache/ are git-ignored
- Open documentation inconsistency: README.md states a primary snapshot of 2026-07-30 while table_12 lists verified caches dated 2026-08-14. Worth reconciling before circulation

How much the partition moves under reasonable alternatives

Agreement with the primary partition ranges from 0.58 to 0.82 ARI, so the boundary is method-dependent even though the direction is not.

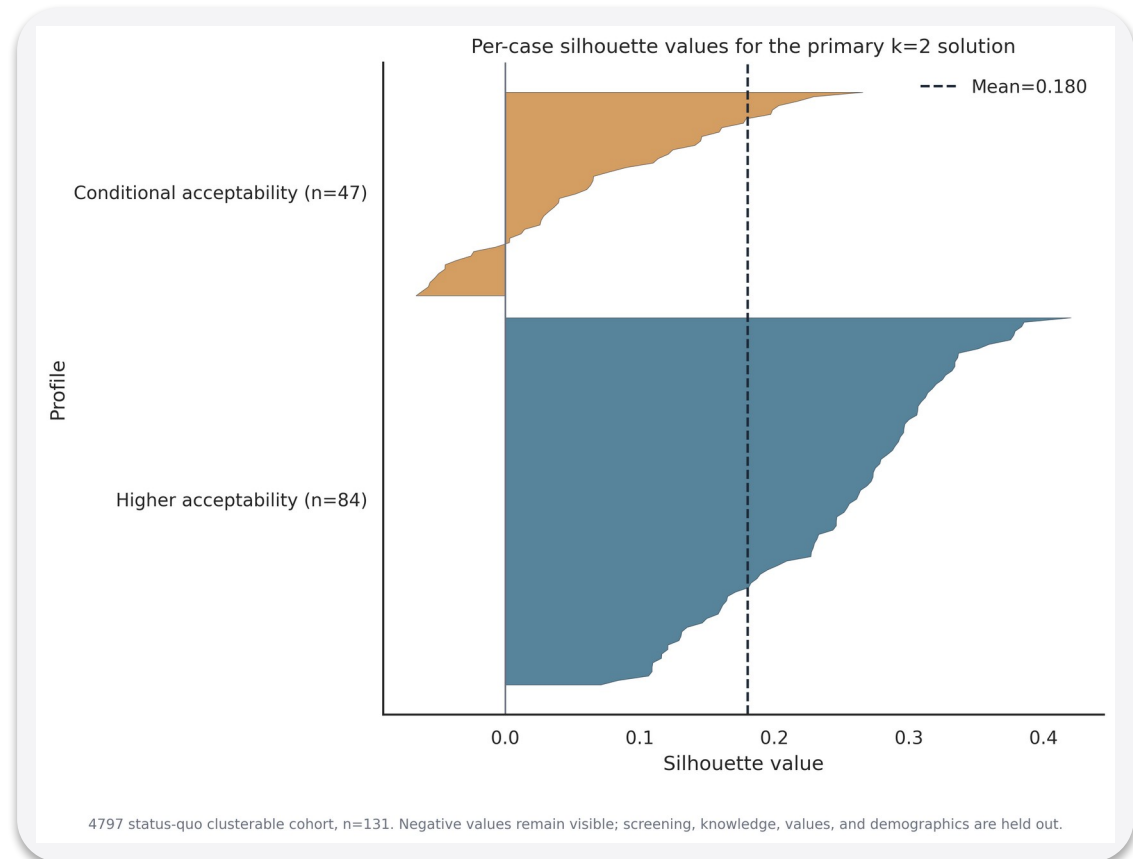
- Complete-domain cases only (n=120): ARI 0.585, silhouette 0.153 — this is the direct test of composite-level median imputation
- Excluding the ethical and moral support domain (n=131): ARI 0.614, and the screening association persists at 62.8% versus 40.8%, a 22.0 pp difference, Fisher $p=0.018$
- 5th to 95th percentile winsorization: ARI 0.690, silhouette 0.165 — marginal outliers are not driving the split
- Ward hierarchical clustering: ARI 0.822, silhouette 0.182, the least method-dependent of the four checks
- Nonlinear embeddings at three UMAP neighbourhood sizes and three t-SNE perplexities show overlapping gradient structure, not separated islands



Model-based checks support two profiles without endorsing them as optimal

The bootstrap likelihood-ratio test rejects a single component, while BIC prefers a three-component diagonal solution.

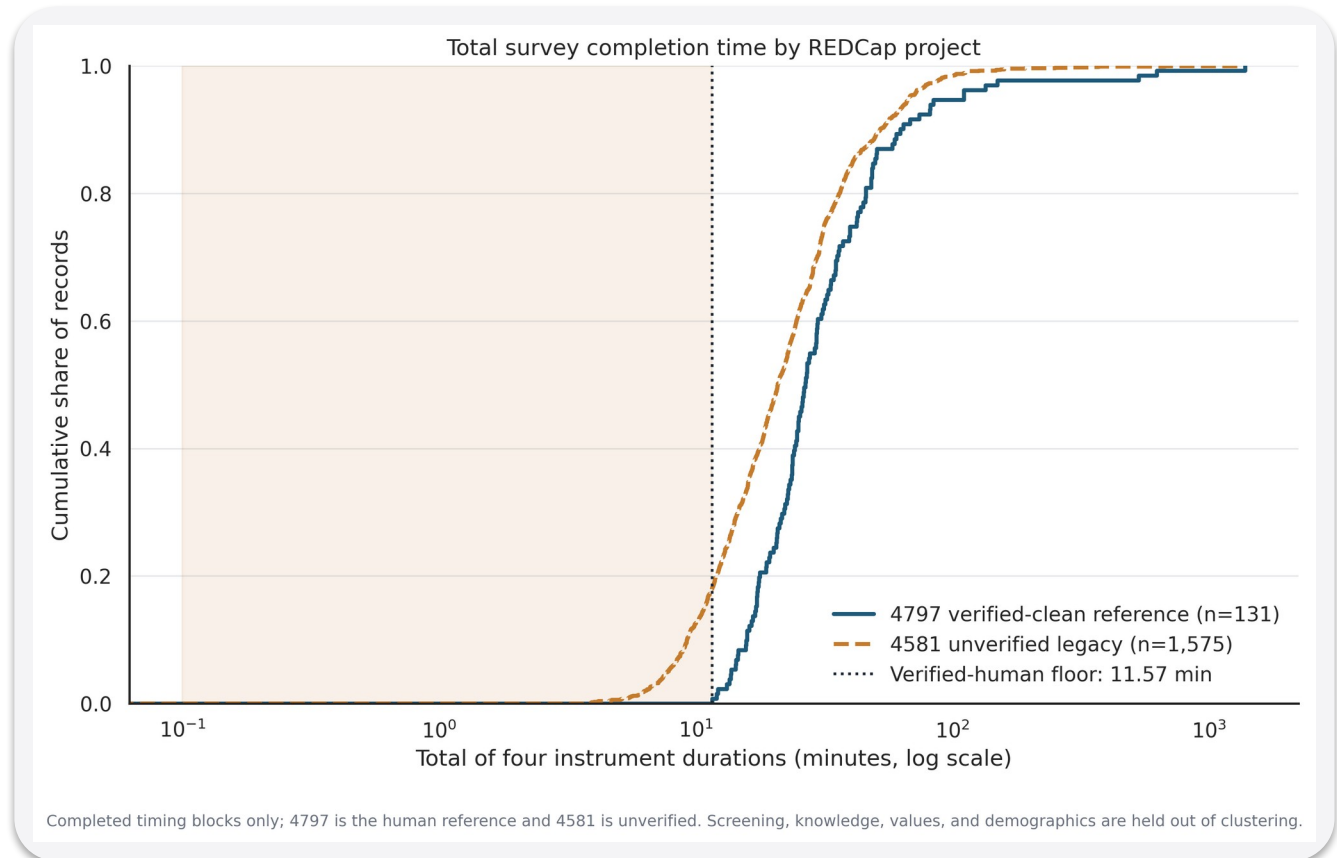
- Bootstrap LRT, $k=2$ versus $k=1$, diagonal covariance: observed likelihood ratio 1587.8 over 100 bootstrap repeats, $p=0.0099$ — one component is rejected
- Best overall BIC is $k=3$ diagonal at 1697.9, against 2329.7 for $k=2$ diagonal, so the information criterion does not endorse two components
- Entropy separation exceeds 0.99 for the retained solution and no record falls below a 0.70 maximum posterior
- Per-case silhouettes are positive for all but a handful of conditional-profile cases; the negative values are left visible rather than trimmed
- Stated plainly: two profiles is the interpretable and stable choice, not the information-criterion optimum



The ten rules, their thresholds, and where the thresholds come from

Every threshold is a percentile or floor computed on the 131 completed 4797 caregivers, so the screen is calibrated to observed human behaviour.

- R1 total time below 11.57 minutes and R2 TFA time below 7.85 minutes, both verified-human minima
- R3 any instrument below the clean 1st percentile; R4 within-block response SD at or below the clean 1st percentile, computed separately for the 1–4, 1–5, 1–6 and 1–7 Likert blocks
- R5 exact ordered Likert fingerprint duplicate at $\geq 80\%$ answered; R6 at least 3 submissions inside a clean-derived 60-second window; R7 near-duplicate open text at TF-IDF cosine ≥ 0.90
- R8 logical and branching inconsistency; R9 impossible demographic combination; R10 4797 instrument-native gate, validation-only and not tier-defining
- Counts over all 1,956 records: R6 1,682, R3 846, R4 589, R2 447, R1 283, R8 44, R7 17, R10 13, R9 2, R5 0



What a branching violation looks like, record by record

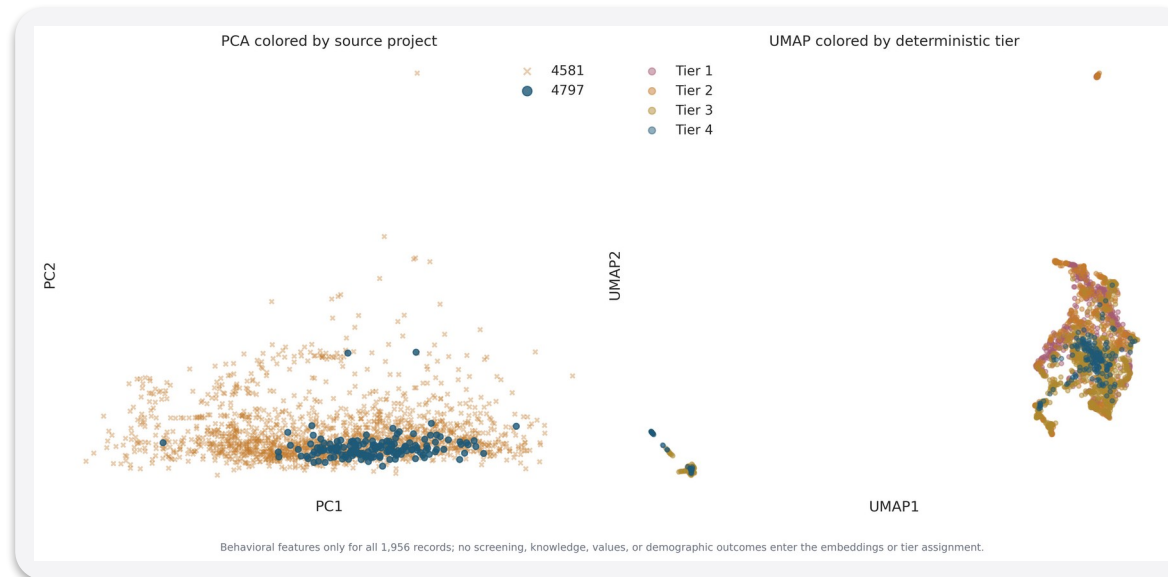
N	VIOLATION FAMILY	PLAIN-LANGUAGE DESCRIPTION	SHARE
16	prenatal_tested_reason_without_prenatal_yes	Reason for prenatal testing answered with no prenatal-yes gate	36.4%
9	child_age_band_count_exceeds_child_count	More distinct child age bands ticked than children reported	20.5%
8	prenatal_testing_mutually_exclusive_pair	Both members of a mutually exclusive prenatal pair selected	18.2%
6	prenatal_not_tested_reason_without_prenatal_no	Reason for not testing answered with no prenatal-no gate	13.6%
3	earlier_diagnosis_no_reason_without_no_gate	No-reason answered without the earlier-diagnosis no gate	6.8%
2	earlier_diagnosis_yes_reason_without_yes_gate	Yes-reason answered without the earlier-diagnosis yes gate	4.5%

Worked example, record 4581_1069: fif_num_children = 2, yet fif_childrens_ages___3, ___4 and ___5 are all ticked. Three distinct age bands for two children is not producible by the form. Every row stores gating_fields, gating_logic, observed_fields, gating_values and observed_values, so any reviewer can re-derive the call without rerunning the pipeline.
Regression fixtures pinned in README.md: sibling-pair collisions 31, 118, 273, 571, 768, 1076, 1554, 1699 · tested-reason orphans 966–979, 981, 1591 · not-tested orphans 364, 416, 987, 1026, 1707, 1741 · earlier-diagnosis orphans 205, 1025, 1137, 1642, 1678.

Why we cannot name a contamination fraction

The estimator assumes labelled positives are drawn at random from the positive class, and our labelled set is simply a different REDCap project.

- Elkan–Noto style nonhuman share in 4581: 0.944 (0.939 to 0.950), with a labelling propensity c of 0.622
- Source-classifier AUC 0.980 measures separability of two projects that differ in recruitment window and in instrument content
- Negative control, random clean half A versus half B: AUC 0.525, close to chance, so the pipeline does not manufacture separation on its own
- Top discriminating features are open-text field count and validation-block time, both instrument-design differences rather than bot signatures
- The prior lab LightGBM artifact was not present in the scoped files, so concordance uses rules, three novelty detectors and the PU classifier only



Limitations we should state before anyone else does

Small n, cross-sectional measurement, an undocumented cohort selection rule, and a trust screen that grades evidence rather than identifying actors.

Power, not equivalence

With 47 and 80 respondents and a 54.3% pooled reference rate, the minimum detectable difference is about 25.6 pp at 80% power. Null characterization results are uninformative about small effects.

Undocumented selection

The 135-ID analytic cohort's selection rule is undocumented in the scoped files, so internal comparisons stand but external generalisation does not follow.

Cross-sectional design

Attitudes and screening intent were measured in the same sitting. No temporal ordering exists, so no causal claim is available in either direction.

Excluded-record tipping point

Reassigning the four cluster-excluded records moves the screening gap between 39.0 and 44.5 pp. The conclusion does not flip anywhere in that range.

Assignment-error correction

BCH-corrected screening rates are 58.3% versus 31.5%, lower than the modal-assignment rates of 70.0% and 27.7% but preserving the ordering.

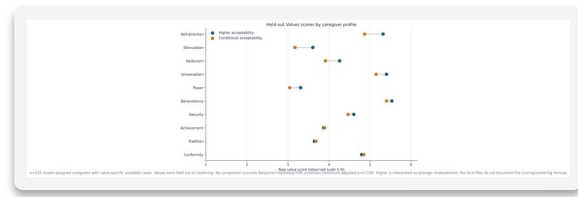
Trust screen is not attribution

Tiers grade how much evidence supports a record. A Tier 1 record may be rushed, malformed or automated, and these data cannot separate those explanations.

Regression harness: 17 of 17 status-quo checks pass within tolerance, including selected_n 135, clusterable_n 131, screen_valid_n 127, silhouette 0.1803, screening gap 42.34 pp and additive cluster OR 8.729. Colour accessibility was checked, with deuteranopia-simulated separation of 0.578 and non-colour backups in place on every figure.

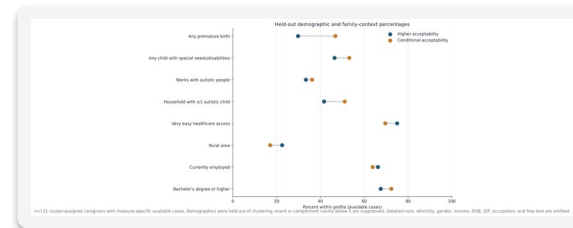
The eight figures not placed earlier, and the claim each one carries

All 21 generated figures now appear in this deck; these eight support claims made on slides 4, 7, 11, 12, A2, A4 and A5.



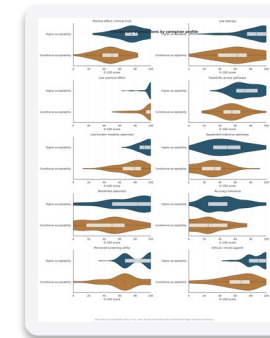
Slide 7 · No value survives FDR control

figure_5_values_characterization.png



Slide 7 · No demographic contrast survives FDR

figure_6_demographic_characterization.png



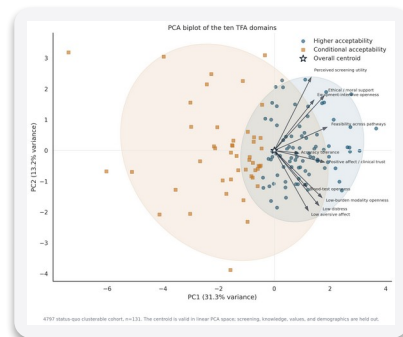
Slide 4 · Distribution shape behind the profile means

figure_17_tfa_domain_distributions.png



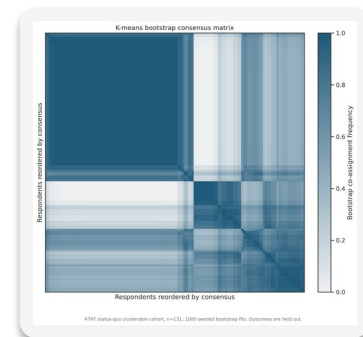
A5 · Legacy response texture versus verified-clean

figure_8_response_fingerprint_heatmap.png



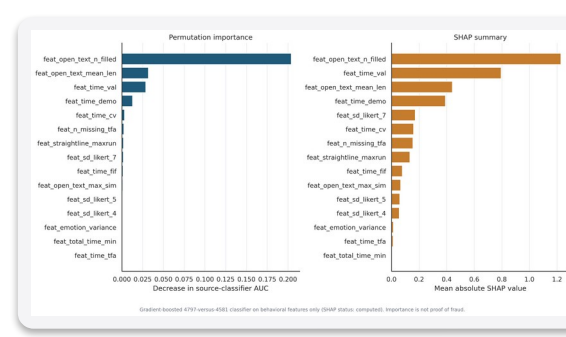
A2 · Linear structure and centroid validity

figure_13_tfa_pca_biplot.png



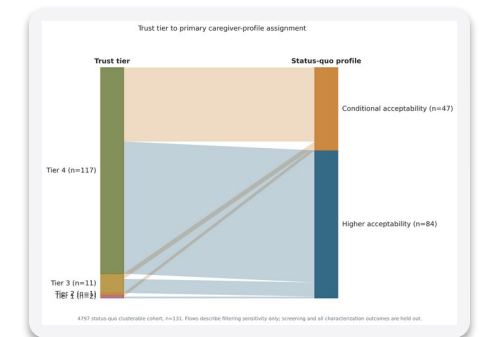
A4 · One solid block, one diffuse block

figure_15_consensus_matrix.png



Slide 12 · Open-text count dominates the PU classifier

figure_11b_pu_feature_importance.png



Slide 11 · Only 14 of 131 sit below Tier 4

figure_19_tier_to_cluster_alluvial.png

Every claim mapped to the file that supports it

SLIDE	CLAIM AS STATED IN THE DECK	SUPPORTING FIGURE AND TABLE FILENAMES
3	k=2 is the most stable and interpretable cluster count; separation is modest	figure_1_cluster_count_diagnostics.png · table_3_cluster_diagnostics.csv
4	Profiles separate most on trust, blood-test openness and ethical support	figure_2_primary_cluster_profiles.png · table_4_cluster_defining_profiles.csv
4	Domain-level distribution shape behind the profile means	figure_17_tfa_domain_distributions.png · table_4_cluster_defining_profiles.csv
5	Definitely-yes screening intent is 70.0% versus 27.7%, a 42.3 pp gap	figure_3_screening_outcome.png · table_5_screening_outcome.csv
6	Adjusted ORs: profile 8.73, ASD at home 4.32, interaction 2.02 (n.s.)	figure_18_logistic_forest.png · table_11_asd_family_context.csv · table_22_logistic_models.csv
7	Verified autism knowledge does not separate the profiles (p=0.099)	figure_4_autism_knowledge.png · table_6_knowledge_summary.csv · table_6b_knowledge_item_results.csv
7	No value survives FDR control; self-direction is the largest raw signal	figure_5_values_characterization.png · table_7_values_summary.csv
7	No demographic contrast reaches FDR $p < 0.56$	figure_6_demographic_characterization.png · table_8_demographics_summary.csv
7	Minimum detectable difference is about 25.6 pp at 80% power	table_29_power_precision.csv
8	Rule co-occurrence and verified-human false-positive rates	figure_9_rule_cooccurrence.png · table_15_rule_false_positive_rates.csv · table_14_fraud_rule_definitions.csv
8	Rule counts across both projects, R6 = 1,682 and R5 = 0	table_15b_rule_counts_by_project.csv
9	Tier 4 pass share is 86.4% in clean_4797 and 3.9% in dirty_4581	table_16_trust_tier_counts.csv (chart built from these values)
10	44 branching and logic impossibilities across six families, all in dirty_4581	table_34_branching_logic_audit.csv · table_15b_rule_counts_by_project.csv
10	R8 verified-human false-positive rate is 0.0% in 131 records	table_15_rule_false_positive_rates.csv

Every claim mapped to the file that supports it

SLIDE	CLAIM AS STATED IN THE DECK	SUPPORTING FIGURE AND TABLE FILENAMES
11	Screening gap holds at 42.3 to 52.2 pp across five inclusion definitions	figure_20_tier_sensitivity.png · table_20_tier_sensitivity.csv
11	Only 14 of 131 clusterable caregivers sit below Tier 4	figure_19_tier_to_cluster_alluvial.png · table_16_trust_tier_counts.csv
12	Maximum detector agreement is Cohen's kappa 0.29	figure_11_detector_scores_and_agreement.png · table_19_detector_agreement.csv
12	Top PU feature is open-text field count (permutation AUC drop 0.204)	figure_11b_pu_feature_importance.png · table_18_detector_feature_importance.csv
13	Five decisions, four implemented, contamination fraction left open	table_30_decision_summary.csv · table_17_contamination_identifiability.csv
A1	Hashed dated caches, halt gates, and field alignment across projects	table_12 · table_13 · table_13b · table_1 · table_31 · README.md
A2	What is clustered: ten standardized TFA domains, softly structured	figure_12_tfa_clustermap.png · table_25_latent_profile_quality.csv
A2	Linear structure and centroid validity in PCA space	figure_13_tfa_pca_biplot.png
A3	Partition agreement ranges 0.585 to 0.822 ARI under alternatives	figure_14_umap_tsne_sensitivity.png · table_10_sensitivity_checks.csv
A4	Bootstrap LRT $p=0.0099$ for $k=2$ vs $k=1$; BIC prefers $k=3$ diagonal	table_24_gmm_bootstrap_lrt.csv · table_23_gmm_diagnostics.csv · figure_15_consensus_matrix.png
A4	Negative per-case silhouettes retained rather than trimmed	figure_16_case_silhouette.png
A5	Timing floors are verified-human minima, not percentiles of pooled data	figure_7_timing_ecdf.png · table_14_fraud_rule_definitions.csv
A5	Legacy response texture differs from the verified-clean reference	figure_8_response_fingerprint_heatmap.png
A7	Verified humans sit inside the legacy cloud; tiers form no islands	figure_10_behavioral_feature_space.png · table_17_contamination_identifiability.csv
A8	Tipping point 39.0–44.5 pp; BCH 58.3% vs 31.5%; 17/17 regression checks pass	table_28 · table_26 · table_27 · table_32_color_accessibility_check.csv

Source: All files resolve under Caregiver Outputs/ unless noted; output_manifest.csv carries a SHA-256 prefix for each

Claims to avoid, and the wording that the evidence will carry

TEMPTING CLAIM	STATUS	SAFER WORDING
Two distinct caregiver types exist	Weak	A two-profile partition of a continuous acceptability gradient is the most stable and interpretable summary of these data.
Higher acceptability leads to screening uptake	Unsupported	Profile membership is strongly associated with held-out screening intent. The design is cross-sectional, so no ordering is available.
94.4% of legacy records are bots	Unsupported	A source classifier separates the two projects at AUC 0.980. The implied separation share is not an identifiable contamination estimate.
The trust screen detects bots	Unsupported	The trust screen grades how much evidence supports each record. It does not identify automated respondents.
Tier 1 records were submitted by bots	Unsupported	Tier 1 records contain hard logical or timing impossibilities under our rules, whatever produced them.
The profiles do not differ demographically	Weak	No demographic difference was detectable here. Differences below roughly 25 percentage points would not be visible at these sample sizes.
Self-direction values distinguish the profiles	Weak	Self-direction shows the largest raw difference among ten values tested but does not survive false-discovery-rate control.
The 4581 subset replicates the finding	Weak	The legacy Tier-4 subset is directionally consistent, with an interval (5.7 to 74.1 pp) too wide to be confirmatory.
R8 implies about 2.5% contamination	Unsupported	R8 confirms that 44 records contain logically impossible responses. Its recall is unknown, so it bounds nothing.
Cluster membership is stable	Weak	The direction of the result is stable. Individual assignments move under alternative preprocessing, ARI 0.585 to 0.822.

Source: Status assigned from table_17, table_20, table_23, table_24, table_29, table_10, table_7, table_8, table_19 and table_34

What was checked before this deck was written

The build script re-reads every source CSV and asserts each anchor; it refuses to write the file on any mismatch.

18 anchors asserted at build time

Combined records 1,956 · clean_4797 177 · dirty_4581 1,779 · R8 44 / 0 / 44 · R8 false positives 0.0% in 131 · all six branching families · dirty tiers 1–3 at 96.1%. All 18 passed.

17 of 17 pipeline checks pass

table_27 confirms selected_n 135, clusterable_n 131, screen_valid_n 127, silhouette 0.1803, screening gap 42.34 pp and additive cluster OR 8.729, each inside its tolerance.

Figures used unmodified

Every figure is the 300-DPI PNG from Caregiver Outputs, uncropped and unedited. The two native charts are drawn from table_16 and table_34 values only.

- Open item: README.md states a primary cache snapshot of 2026-07-30 while table_12 lists verified caches dated 2026-08-14. The deck follows the CSV. Reconcile before circulation
- Open item: the selection rule producing the 135-ID analytic cohort is undocumented in the scoped files, which limits external generalisation claims
- Known gaps carried forward: the prior lab LightGBM artifact was not present in the scoped files, and R10's email anti-fraud component is unavailable under current API permissions
- Not verified here: whether the figure PNGs were regenerated from the current pipeline state. output_manifest.csv carries a SHA-256 prefix per artifact, which would settle it